

Feed Rate Effect of Micro-milling Process on Cut Surface Quality of I-Plate Implants

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ABSTRACT

Micro-milling of commercially pure titanium is strongly affected by size effects, ductility, and material adhesion, which can influence dimensional accuracy, burr formation, and surface composition. This study investigated the effect of feed rate from 10 to 90 mm/min on the machining quality of I-Plate implants. Machining was performed at 18,000 rpm with a depth of cut of 30 μm per pass under dry conditions using a 1 mm carbide end mill. Dimensional accuracy was evaluated on inner, outer, and straight profiles, while burr height and surface composition were also measured. Surface composition was examined by scanning electron microscopy coupled with energy-dispersive X-ray spectroscopy. Mean dimensional accuracy increased from 98.0% to 99.1%, while mean burr height decreased from 69.9 to 40.3 μm . The initial material contained 99.64 wt.% titanium. Titanium reduction was greatest at 20 mm/min and remained lower from 30 to 90 mm/min, while iron enrichment was most pronounced at low feed rates. The smallest titanium reduction occurred at 50 mm/min. Overall, 90 mm/min provided the best geometric and edge quality with relatively low compositional change.

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1. INTRODUCTION

Micro-milling is a material removal process that uses microscale cutting tools to produce small three-dimensional features with high geometric flexibility. Unlike conventional milling, however, cutting mechanics at the microscale cannot be scaled down linearly. The cutting-edge radius, undeformed chip thickness, tool run-out, and tool deflection become comparable to the dimensions

of the machined features. This condition gives rise to the size effect, in which rubbing and ploughing become increasingly significant when the undeformed chip thickness is insufficient for effective chip formation by shearing [1,2]. Titanium and its alloys are widely used for biomedical components because of their corrosion resistance, biocompatibility, and mechanical properties [3]. From a manufacturing perspective, however, titanium is relatively

difficult to machine. In the micro-milling of commercially pure titanium (CP-Ti), a high ratio of cutting-edge radius to undeformed chip thickness can promote ploughing, plastic deformation, built-up material formation, and adhesive wear [4,5]. Consequently, machining quality cannot be assessed solely by the successful formation of the intended geometry. Dimensional accuracy, edge quality, and the condition of the machined surface must also be considered.

Bone fixation plates are an example of components that require small geometries, fixation holes, and precise external contours. Computer-aided design and manufacturing combined with CNC milling have been applied to the fabrication of mandibular reconstruction plates based on predefined geometries [6]. Die-sinking electrical discharge machining has also been investigated for the manufacture of CP-Ti microplates, with product quality evaluated in terms of dimensional accuracy, surface roughness, white-layer characteristics, hardness, and processing time [7,8]. These studies demonstrate that both the manufacturing process and its operating parameters directly influence the final quality of microplate components.

Burr formation is one of the most important edge defects in micro-milling. Burrs are produced when material undergoes plastic deformation and flows toward the sides of the cutting path before being completely separated as a chip. At the microscale, burr dimensions may become comparable to the dimensions of the machined features, making subsequent deburring capable of altering component geometry [9, 10]. Studies on CP-Ti have shown that feed per tooth, the ratio of feed to cutting-edge radius, depth of cut, and toolpath strategy affect burr formation [5,11]. Feed rate should therefore be evaluated not only in terms of productivity but also in terms of its ability to preserve the intended geometry and suppress burr formation.

Dimensional accuracy is another important response because inner, outer, and straight profiles involve different toolpath kinematics and

tool engagement conditions. Cutting conditions and tool geometry can produce deviations from the intended surface topography or nominal dimensions during micro-milling [12]. Previous studies on the micro-milling of CP-Ti implant microplates have also demonstrated that process parameters affect dimensional accuracy. However, these studies employed combinations of machining parameters within relatively low feed-rate ranges and focused primarily on dimensional accuracy [13]. Isolating feed rate over a wider range may therefore provide additional insight into the geometric response of I-Plate features.

In addition to geometric quality, changes in surface composition after machining are relevant for materials intended for implant applications. X-ray fluorescence (XRF) has been widely used to identify adhered material on cutting tools and localized compositional changes associated with machining. Material adhesion and built-up edge formation are commonly reported tool-wear mechanisms in the micro-milling of titanium [14–16]. Nevertheless, the titanium micro-milling literature has predominantly focused on burr formation, surface roughness, cutting forces, tool wear, and surface integrity. Integrated evaluation of multi-profile dimensional accuracy, burr height, and compositional changes on the machined surface of CP-Ti microplate geometries remains limited.

Accordingly, this study investigates the effect of feed rate on three quality characteristics of micro-milled CP-Ti I-Plate implants: dimensional accuracy, burr height, and surface compositional changes used as an indicator of contamination. Feed rates ranging from 10 to 90 mm/min were investigated while spindle speed, depth of cut, cutting tool, and machining conditions were kept constant. Used cutting tools were also examined to provide supporting evidence of possible adhesion or material transfer during machining. This study does not assess the biological safety or toxicological implications of the implants. In this work, the term *contamination* is used operationally to describe measurable changes in elemental composition on the machined surface. measurements. The average composition is presented in Table 1. The Ti fraction has a mean value of 99.74 wt.%. These measurements were used as the experimental baseline for evaluating compositional changes after machining.

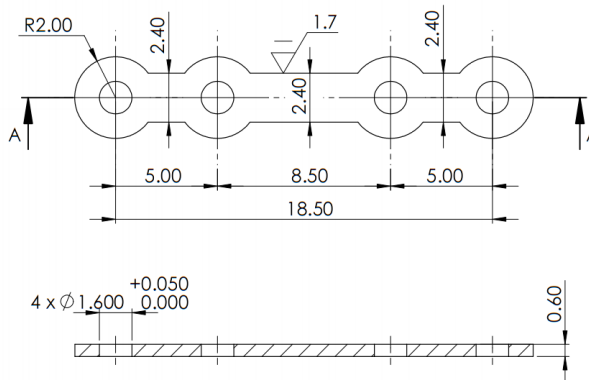
2. MATERIALS AND METHODS

The workpiece material was a 0.6 mm thick plate of Commercially Pure Titanium (CP-Ti). The elemental composition of the initial material was determined from X-ray fluorescence (XRF)

Table 1. Chemical composition of raw material.

Element	Chemical composition [%]				
	Ti	V	Cr	Fe	Pd
Weight (wt%)	99.74	0.06	0.07	0.12	0.01

The I-Plate geometry consisted of four collinear holes, outer circular profiles surrounding the hole regions, and two straight connecting sections. The nominal dimensions were 18.50 mm in overall length, 1.60 mm in hole diameter, 2.00 mm in outer-contour radius, and 2.40 mm in straight-section width. This geometry was selected because a single specimen incorporates three distinct profile characteristics that can be evaluated: inner, outer, and straight profiles. The CAD model was created using Autodesk Inventor and served as the reference for the nominal dimensions and toolpath generation.

SECTION A-A
SCALE 5:1**Fig. 1.** Implant I-Plate.

The I-Plate model was processed using Aspire 12.5 to generate the machining toolpaths. Four pocket toolpaths were used to produce the holes, followed by one profile toolpath to machine the outer contour. The total programmed cutting depth was 0.66 mm to ensure complete penetration through the 0.60 mm thick plate. The depth of cut was set to 0.03 mm, or 30 μm , per pass, resulting in 22 passes. The machining program was executed on a CNC router using Mach3 software. The workpiece was secured on an acrylic fixture that provided clearance beneath the cutting region for full tool penetration. A four-flute carbide end mill with a diameter of 1 mm was used. The spindle speed was maintained at 18,000 rpm, and all experiments were conducted under dry-cutting conditions.

Feed rate was the independent variable and was varied from 10 to 90 mm/min in increments of 10 mm/min. With four cutting edges and a spindle speed of 18,000 rpm, feed per tooth was calculated using Eq. (1).

$$f_z = \frac{V_f}{n \cdot z} \quad (1)$$

Under these conditions, the feed per tooth increased from 0.139 $\mu\text{m}/\text{tooth}$ at 10 mm/min to 1.250 $\mu\text{m}/\text{tooth}$ at 90 mm/min. This range was selected to examine changes in machining response as the chip load per tooth increased. A numerical minimum chip thickness was not specified because the actual cutting-edge radius of the tool was not measured in this study.

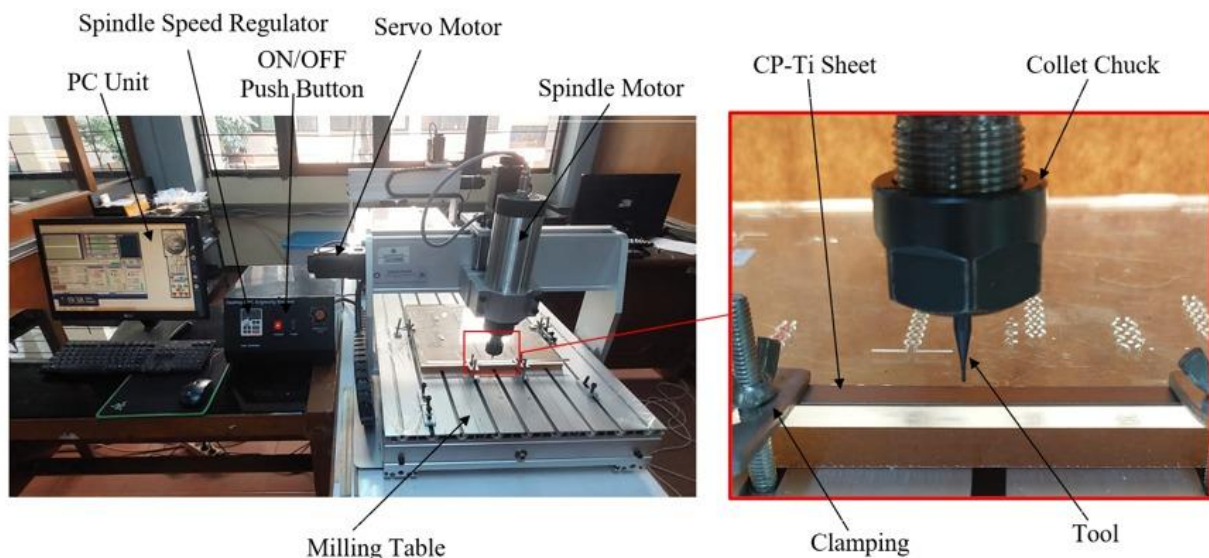
**Fig. 2.** CNC-Router and tool configuration

Table 2. Cutting Data

Parameter	Value
Material	CP-Ti plate 0,60 mm
Tool	Carbide endmill
Tool diameter	Ø1 mm
Number of flutes	4
Spindle speed	18.000 rpm
Cutting speed	56,5 m/min
Feed rate	10–90 mm/min
Feed per tooth	0.139–1.250 µm/tooth
Depth of cut per pass	30 µm
Total depth of cut	0,66 mm
Cutting system	Dry cutting
Toolpath	4 pockets, 1 profile

Dimensional accuracy was measured using a Dino-Lite digital microscope, which was calibrated before measurement and operated using DinoCapture 3.0. Measurements were performed at 50× magnification. The acquired images were subsequently processed using ImageJ to assist with edge identification and dimensional measurement. Three geometric characteristics were evaluated: hole diameter (*inner*), outer circular contour diameter or size (*outer*), and straight-section width (*straight*). Dimensional accuracy was calculated relative to the nominal CAD dimensions using Eq. (2).

$$DA = \left(\frac{1 - |D_{measured} - D_{nominal}|}{D_{nominal}} \right) \times 100\% \quad (2)$$

Burr height was measured at 120× magnification with the specimen positioned laterally so that material protruding from the machined edge could be observed relative to the reference surface. The measurements were classified according to the straight and outer profiles. For each feed rate, the mean burr height was calculated as the average of the two profile categories using Eq. (3):

$$H_{mean} = \frac{H_{straight} + H_{outer}}{2} \quad (3)$$

Elemental composition was analyzed by SEM-EDS on the initial material, micro-milled specimens, and used cutting tools. The acquisition time for each measurement was 15 s. The initial material was analyzed at four

locations. For each feed rate, one machined I-Plate specimen was measured at three different surface locations. These measurements represented spatial replicates on the same specimen rather than independent machining replicates. The results were therefore treated descriptively and averaged to represent the surface composition corresponding to each feed-rate condition.

The principal compositional change was evaluated from the reduction in Ti fraction relative to the baseline composition of the initial material. The Ti reduction indicator, ΔTi , was calculated using Eq. (4), while enrichment of element i was calculated using Eq. (5). The ΔTi values were expressed in percentage points (pp), rather than as an absolute mass fraction of contamination. This approach avoids assuming that all elements other than Ti originated from external contamination, because minor elements were also present in the initial material.

$$\Delta Ti = \bar{X}_{Ti, initial} - \bar{X}_{Ti, specimen} \quad (4)$$

$$\Delta X_i = \bar{X}_{Ti, specimen} - \bar{X}_{Ti, initial} \quad (5)$$

Two used cutting tools from the experimental series were also analyzed. Each tool was measured at three locations within the cutting region and at one location on the shank. The two tools were treated as equivalent observations of used cutting tools and were not associated with specific feed-rate conditions. Comparison between the cutting regions and the shanks was used as supporting evidence for Ti enrichment on tool regions subjected to machining contact.

3. MATERIALS AND METHODS

3.1 Feed Rate Effect on Dimensional Accuracy

Fig. 3 show an overall improvement in dimensional accuracy with increasing feed rate. Mean dimensional accuracy increased from 98.0% at 10 mm/min to 99.1% at 90 mm/min. For the individual profiles, the accuracy of the inner profile increased from 97.1 to 98.5%, the outer profile from 98.0 to 99.2%, and the straight profile from 98.8 to 99.7%. Although the trend was not strictly monotonic at every feed rate, particularly for the inner and outer profiles, the overall results indicate decreasing deviation from the nominal geometry as feed rate increased. This tendency can be interpreted in relation to the increase in feed per tooth. At constant spindle

speed, f_z increased from 0.139 to 1.250 $\mu\text{m}/\text{tooth}$ over the investigated feed-rate range. At very low f_z , the resulting undeformed chip thickness is more likely to be small relative to the cutting-edge radius, and the tool therefore departs from the ideal behavior of a perfectly sharp cutting edge. Under such conditions, elastic-plastic deformation and ploughing can become more pronounced, while material removal by shearing becomes less dominant [1,4]. Elastic recovery after tool passage may consequently contribute to dimensional deviation. As f_z increases, the contribution of effective chip formation may increase, allowing the machined geometry to follow the nominal toolpath more closely.

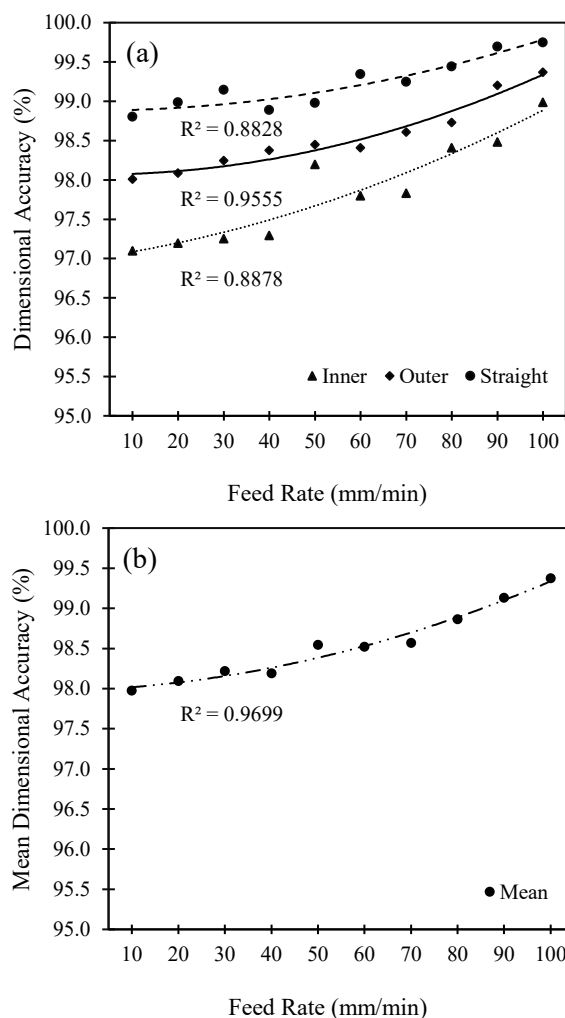


Fig. 3. Dimensional accuracy of specimens.

The straight profile maintained the highest dimensional accuracy throughout the investigated range, whereas the inner profile consistently showed the lowest values. This difference indicates that toolpath geometry also influences dimensional response. Along curved

paths, changes in cutting direction and tool engagement may alter the effective lateral loading and cutting-edge position compared with a straight toolpath. Previous micro-milling studies have similarly shown that cutting parameters and tool geometry can affect deviations from the intended geometry [12].

The present results should also be considered in relation to previous work on CP-Ti microplates using a Taguchi design with multiple machining parameters and feed rates of 5–15 mm/min [13]. Differences in parameter range, parameter interactions, and cutting conditions prevent a direct one-to-one comparison of the feed-rate response between that multifactorial study and the present single-variable experiment. Under the tool, spindle-speed, and depth-of-cut conditions used here, however, increasing the feed rate from 10 to 90 mm/min generally shifted the process toward conditions that produced better dimensional accuracy.

3.2 Effect of Feed Rate on Burr Height

Burr height showed an opposite overall trend to dimensional accuracy. Mean burr height decreased from 69.9 μm at 10 mm/min to 40.3 μm at 90 mm/min, corresponding to a reduction of 42.3%. Burr height on the straight profile decreased from 37.8 to 19.7 μm , while that on the outer profile decreased from 102.0 to 61.0 μm . A local increase occurred at 70 mm/min, indicating that the relationship between feed rate and burr height was not strictly monotonic. Nevertheless, both profiles showed a clear overall reduction in burr height from the lower to the higher feed-rate conditions.

Burr formation in titanium micro-milling is closely associated with the size effect and minimum chip thickness. When the undeformed chip thickness is very small relative to the cutting-edge radius, ductile material may be displaced and plastically deformed rather than efficiently removed as a chip. Plastic side flow toward the machined edge can consequently produce larger burrs. Increasing f_z increases the nominal chip load per tooth and may enhance the contribution of shearing, thereby reducing the amount of material displaced toward the edge [4,5,11]. This interpretation is consistent with previous CP-Ti studies showing that the ratio of feed per tooth to cutting-edge radius and the toolpath strategy influence burr formation [11].

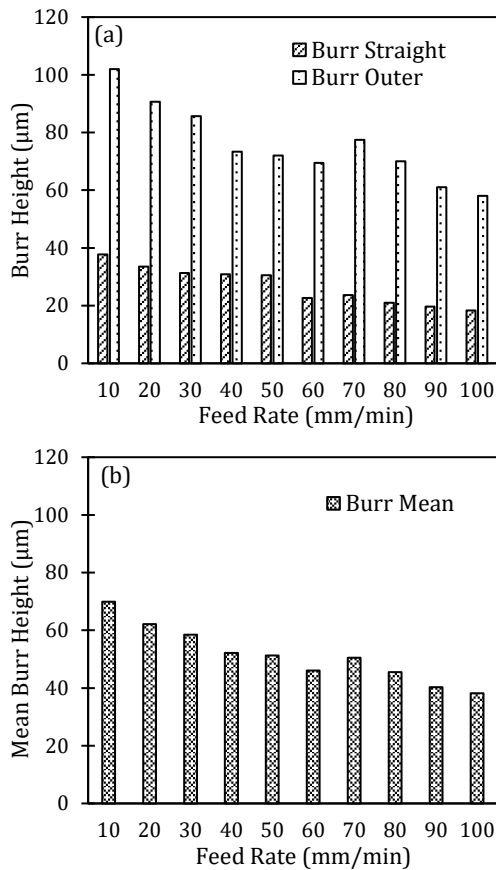


Fig. 4. Measured burr heights of specimens.

Burr height on the outer profile was consistently greater than that on the straight profile. This difference suggests that toolpath geometry influences the burr-formation mechanism. Along the outer contour, the direction of tool engagement and cutting-edge exit conditions vary continuously, which may produce non-

uniform deformation at the machined edge. In contrast, the straight profile has a more constant feed direction. Because cutting forces, tool run-out, and the actual cutting-edge radius were not measured, this explanation should be regarded as a mechanistic interpretation rather than direct verification of a single causal mechanism. From a practical perspective, the observed trend indicates that increasing feed rate within the investigated range simultaneously improved dimensional accuracy and reduced burr formation.

3.3 Changes in Elemental Composition and Contamination Indicators

The independently measured raw-material XRF composition was used as the reference for evaluating compositional changes after micro-milling. The raw material contained 99.74 wt.% Ti, 0.06 wt.% V, 0.07 wt.% Cr, 0.12 wt.% Fe, and 0.01 wt.% Pd. Al was absent from the reported raw-material composition and was therefore assigned a baseline value of 0.00 wt.% for the difference calculation. For each feed rate, the values reported below are the means of three spatial XRF measurements on one machined I-Plate specimen. These measurements represent spatial replicates rather than independent machining replicates; consequently, the compositional results are interpreted descriptively.

Table 3. Mean elemental composition of micro-milled specimens measured by XRF (wt.%).

Feed rate (mm/min)	Chemical Composition (wt.%)					
	Ti	Fe	Al	V	Cr	Pd
10	97.18	1.63	0.73	0.21	0.06	0.05
20	96.16	3.02	0.60	0.00	0.08	0.04
30	98.47	0.38	0.77	0.11	0.11	0.04
40	98.76	0.25	0.61	0.20	0.05	0.03
50	99.02	0.21	0.48	0.00	0.18	0.03
60	98.79	0.24	0.66	0.10	0.12	0.03
70	98.66	0.20	0.72	0.29	0.02	0.03
80	98.76	0.30	0.66	0.00	0.11	0.04
90	98.79	0.26	0.69	0.02	0.12	0.02

Table 4. Change in mean elemental composition relative to the raw-material XRF baseline (wt.%).

Feed rate (mm/min)	Chemical Composition (wt.%)					
	ΔTi	ΔFe	ΔAl	ΔV	ΔCr	ΔPd
10	-2.56	+1.51	+0.73	+0.15	-0.01	+0.04
20	-3.58	+2.90	+0.60	-0.06	+0.01	+0.03
30	-1.27	+0.26	+0.77	+0.05	+0.04	+0.03
40	-0.98	+0.13	+0.61	+0.14	-0.02	+0.02
50	-0.72	+0.09	+0.48	-0.06	+0.11	+0.02
60	-0.95	+0.12	+0.66	+0.04	+0.05	+0.02
70	-1.08	+0.08	+0.72	+0.23	-0.05	+0.02
80	-0.98	+0.18	+0.66	-0.06	+0.04	+0.03
90	-0.95	+0.14	+0.69	-0.04	+0.05	+0.01

All machined specimens exhibited a lower Ti fraction than the raw-material baseline. At 10 and 20 mm/min, the mean Ti fractions were 97.18 and 96.16 wt.%, corresponding to ΔTi values of -2.56 and -3.58 percentage points, respectively. At 30-90 mm/min, the Ti fraction remained between 98.47 and 99.02 wt.%, with ΔTi values between -1.27 and -0.72 percentage points. The largest Ti deviation therefore occurred at 20 mm/min, whereas the smallest deviation occurred at 50 mm/min. Although the relationship was not monotonic, the two lowest feed rates formed a distinct group with substantially larger Ti reductions than the higher-feed conditions.

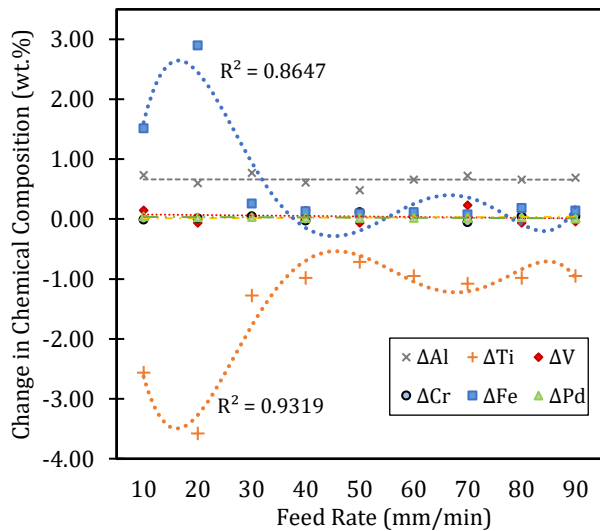


Fig. 5. Change in chemical composition of specimens.

Fe showed the clearest enrichment at low feed rates. Relative to the raw-material value of 0.12 wt.%, the specimens machined at 10 and 20 mm/min contained 1.63 and 3.02 wt.% Fe, corresponding to increases of +1.51 and +2.90 percentage points. By contrast, Fe remained between 0.20 and 0.38 wt.% at 30-90 mm/min, equivalent to increases of only +0.08 to +0.26 percentage points. Al exhibited a different behavior. Although Al was absent from the raw-material composition, it was detected in every machined specimen at 0.48-0.77 wt.%. The lowest Al fraction was measured at 50 mm/min, while the highest occurred at 30 mm/min; therefore, Al enrichment did not show a monotonic relationship with feed rate.

The remaining elements showed smaller and less systematic variations. V changed by -0.06 to +0.23 percentage points, Cr by -0.05 to +0.11 percentage points, and Pd by +0.01 to +0.04 percentage points relative to the raw material. These fluctuations were considerably smaller than the low-feed Fe enrichment and the associated decrease in Ti fraction. Because XRF reports relative elemental fractions, a lower Ti wt.% should not be interpreted as direct chemical loss of Ti from the workpiece. Instead, it primarily indicates an increased relative contribution of other detected elements in the analyzed region. Accordingly, ΔTi is used here as a compositional-change indicator rather than as an absolute mass fraction of contamination.

The larger compositional deviations at 10-20 mm/min are phenomenologically consistent with

the size-effect regime of titanium micro-milling. At very small chip loads, increased rubbing and ploughing can increase tool-workpiece contact and promote adhesion, smearing, and retention or transfer of material [14-16]. The XRF results support the presence of machining-induced compositional change, but they do not independently identify the source of every detected element.

3.4 Evidence of Material Transfer at the Cutting Tool

XRF measurements of two end mills provide complementary evidence of material transfer at the tool-workpiece interface. For Endmill-1, which is a brand-new tool, the mean Ti fraction from three measurements in the cutting region was 30.82 wt. For Endmill-2, which was used for machining of the specimens, the corresponding values were 39.41 wt.% in the cutting region.

Table 5. Measured chemical composition of tools.

Tool	Chemical composition [%]			
	Ti	W	Co	Fe
Endmill-1	30.82	54.94	7.00	0.12
Endmill-2	39.41	50.57	6.61	0.06

The consistently higher Ti fraction in the cutting regions is compatible with adhesion of CP-Ti workpiece material to the tool during machining. At the same time, the relative W fraction became lower in the cutting regions after machining. This difference should not be interpreted as an equivalent absolute loss of W because the reported XRF fractions are compositionally normalized; an increase in adhered Ti and other constituents necessarily reduces the relative contribution of the underlying tool elements in the measured composition.

Taken together with the Al enrichment measured on the machined specimens, the tool results are consistent with material transfer in both directions across the cutting interface: Ti-rich workpiece material can adhere to the cutting region, while Al-containing material from the tool coating can be transferred to the workpiece. Such adhesion and built-up material are widely reported in micro-machining of titanium [14-16]. The tool data are therefore treated as supporting evidence of transfer and adhesion during

machining, not as a basis for quantifying the effect of feed rate on tool contamination or wear.

Considering the three investigated responses together, no single feed rate produced the optimum value for every criterion. A feed rate of 90 mm/min provided the highest dimensional accuracy and lowest burr height, while 50 mm/min produced the smallest compositional deviation from the raw material. Feed rates from 30 to 90 mm/min generally showed a more favorable balance than 10–20 mm/min, particularly because the latter exhibited substantially greater compositional changes. Since no multi-objective optimization or response weighting was performed, these conditions are interpreted according to the individual measured responses rather than as a universal optimum.

4. CONCLUSION

The effect of feed rate on the micro-milling quality of CP-Ti I-Plate specimens was evaluated in terms of dimensional accuracy, burr height, and elemental composition. Within the investigated range of 10–90 mm/min, increasing feed rate generally improved geometric quality. Mean dimensional accuracy increased from 98.0% at 10 mm/min to 99.1% at 90 mm/min, while mean burr height decreased from 69.9 μm to 40.3 μm . The straight profile consistently showed the highest dimensional accuracy and lower burr height than the outer profile, indicating that toolpath geometry also influenced the machining response.

XRF analysis showed that compositional changes were greatest at the two lowest feed rates. Relative to the raw-material composition of 99.74 wt.% Ti, the largest Ti deviation occurred at 20 mm/min (–3.58 percentage points), whereas the smallest occurred at 50 mm/min (–0.72 percentage points). Fe enrichment was also most pronounced at low feed rates, while Al was detected on all machined specimens despite being absent from the raw-material composition. Together with the higher Ti fraction measured on the used cutting tool, these results provide supporting evidence of material transfer across the tool–workpiece interface.

Overall, 90 mm/min provided the best dimensional accuracy and lowest burr height, whereas 50 mm/min produced the smallest compositional deviation. Feed rates from 30 to 90

mm/min therefore provided a more favorable overall machining response than 10–20 mm/min. These findings demonstrate that feed-rate selection in CP-Ti micro-milling should consider geometric accuracy, burr formation, and machining-induced compositional changes simultaneously rather than relying on a single quality criterion.

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